

Computer Exercise 3: Linear Mixed Models

Lars Ronnegard

May 15, 2019

Introduction

The data set `estrone.data` contains 80 observations of estrone levels (y) from 5 females (from the book: Pawitan 2001 In all likelihood).

```
y <- c(13.61728, 13.61728, 13.42423, 13.01030, 13.97940, 13.42423, 14.31364,
      13.97940, 13.42423, 13.42423, 13.61728, 13.61728, 14.31364, 12.78754,
      13.61728, 12.55273, 13.97940, 15.18514, 14.31364, 14.31364, 14.77121,
      14.47158, 13.80211, 13.42423, 14.14973, 14.77121, 14.77121, 14.62398,
      14.62398, 15.68202, 13.80211, 14.47158, 15.79784, 15.79784, 16.12784,
      15.79784, 15.79784, 15.05150, 15.79784, 16.23249, 15.44068, 16.02060,
      16.12784, 15.68202, 14.47158, 15.56303, 14.77121, 15.68202, 11.46128,
      12.04120, 11.76091, 12.78754, 13.01030, 13.42423, 12.04120, 12.78754,
      12.30449, 12.55273, 13.01030, 12.55273, 10.79181, 12.30449, 11.76091,
      11.13943, 16.62758, 15.56303, 14.77121, 14.62398, 15.56303, 14.91362,
      14.77121, 15.05150, 15.05150, 14.91362, 14.77121, 15.05150, 13.97940,
      14.62398, 14.91362, 15.05150)

id <- c( 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
      2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2,
      3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3,
      4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4,
      5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5 )

#Collect all the data in a data frame
estrone.data <- data.frame(y=y, id=as.factor(id))
```

Exercise 1

Fit a linear mixed model using the `lmer()` function in the `lme4` package with a fixed intercept and female as individual. What is the estimated residual variance? What is the estimated random effect variance?

Exercise 2

Fit the model in Exercise 1 using the `hglm2` function in the `hglm` package. Do you get the same estimates using the `hglm2()` function and the `lmer()` function? Plot the diagnostic plots.

Exercise 3

Construct the design matrix X and the incidence matrix Z in R. Fit the model in Exercise 2 using the `hglm` function in the `hglm` package. Do you still get the same estimates? Plot the diagnostic plots.

Exercise 4 (If you get time)

Compute a 95% confidence interval for the random effect variance.